

Homework 2: Sequence objectives, state, and dependence

Due Friday, October 2, 2026, at 11:59 p.m. This assignment is worth 100 points and 10% of the course grade. Late homework is not accepted; the two-lowest-grade replacement described in the syllabus covers ordinary illness, overload, and missed work. Complete every part listed under Points. Parts not listed there are optional and ungraded. Submit one ZIP archive named hw02-ccid.zip, containing a single top-level directory named hw02-ccid with your PDF, LaTeX source, and any supporting files. Replace ccid with your University of Alberta CCID.

General instructions. Justify every answer, including yes/no answers, even when the question does not explicitly ask for a justification.

Exercise profiles. Filled mountains show difficulty on a four-mountain scale: one is a direct application, two uses a familiar method, three combines ideas, and four is a substantial challenge. Filled cogs show technical load on a three-cog scale: one is light, two is sustained, and three is dense. Filled hourglasses show estimated time: one is under 20 minutes, two is 20–45 minutes, three is 45–90 minutes, and four is more than 90 minutes. Pale pictograms show the unused part of each scale. The work row uses the canonical order  proof,  calculation,  construction,  computation, and  interpretation; pale icons mark work types not used.  marks the intended learning outcome. Time estimates cover the non-optional parts. Ratings are not prerequisites or grading weights.

Name: YOUR NAME Student ID: YOUR STUDENT ID

A missed final-position dependency

Points: (a): 3, (b): 3, (c): 4; total 10.

Exercise 1: A missed dependency at the final position.*

 Show that sequence KL divided by sequence length can vanish while final-answer error stays fixed.

Let $T \geq 2$, let $\mathcal{X} = \{0, 1\}^T$, and let P_{par} and Q_{coin} be probability mass functions on $\mathcal{X}$. Under P_{par} , the first $T - 1$ bits are independent fair coins and the last is their parity: $X_T = (\sum_{t < T} X_t) \bmod 2$. Under Q_{coin} , all T bits are independent fair coins. Define the task cost $c_{\text{par}} : \mathcal{X} \rightarrow \{0, 1\}$ by

$$c_{\text{par}}(x_{1:T}) = \mathbb{I} \left\{ x_T \neq \left(\sum_{t < T} x_t \right) \bmod 2 \right\}.$$

(a) Show that for every $t < T$ and every $x_{<t} \in \{0, 1\}^{t-1}$,

$$P_{\text{par}}(\cdot \mid x_{<t}) = Q_{\text{coin}}(\cdot \mid x_{<t}) = \text{Bernoulli}(1/2).$$

Show also that for every $x_{<T} \in \{0, 1\}^{T-1}$,

$$P_{\text{par}}(\cdot \mid x_{<T}) = \delta_{\text{par}(x_{<T})}, \quad Q_{\text{coin}}(\cdot \mid x_{<T}) = \text{Bernoulli}(1/2),$$

where $\text{par}(x_{<T}) = (\sum_{t < T} x_t) \bmod 2$.

(b) Use the KL chain rule to show that

$$D(P_{\text{par}}, Q_{\text{coin}}) = \log 2, \quad \frac{D(P_{\text{par}}, Q_{\text{coin}})}{T} = \frac{\log 2}{T}.$$

(c) Derive

$$\mathbb{E}_{P_{\text{par}}}[c_{\text{par}}] = 0, \quad \mathbb{E}_{Q_{\text{coin}}}[c_{\text{par}}] = \frac{1}{2},$$

and verify the equality

$$\mathbb{E}_{Q_{\text{coin}}}[c_{\text{par}}] = 1 - e^{-D(P_{\text{par}}, Q_{\text{coin}})}.$$

(d) For this part, let $\mathcal{X}$ be any finite alphabet, and let P, Q be probability distributions on $\mathcal{X}^+ = \cup_{t \geq 1} \mathcal{X}^t$, the set of finite sequences taking values in $\mathcal{X}$. Let $c: \mathcal{X}^+ \rightarrow [0, 1]$. Assume that $\mathbb{E}_P[c] = 0$. For $x \in \mathcal{X}^+$, let $|x|$ denote its length. Assume that $P(\mathcal{X}^T) > 0$ and $Q(\mathcal{X}^T) > 0$ for every $T \geq 1$. For $x \in \mathcal{X}^T$, define the conditional probability mass functions

$$P_T(x) = \frac{P(x)}{P(\mathcal{X}^T)}, \quad Q_T(x) = \frac{Q(x)}{Q(\mathcal{X}^T)}.$$

Show that $\mathbb{E}_{P_T}[c] = 0$ for every $T \geq 1$. Prove or disprove the following statement: If $D(P_T, Q_T)/T \rightarrow 0$ as $T \rightarrow \infty$ then $\lim_{T \rightarrow \infty} \mathbb{E}_{Q_T}[c] = 0$. That is, if the average KL between P_T and Q_T vanishes, the cost under Q_T vanishes.

Solution

Write your solution here.

Points: (a): 8, (b): 12; total 20.**Exercise 2: Fitting parity with independent-bit models.***

🔗 Compare fitting the parity law directly with fitting it using independent-bit models.

Let $T \geq 2$, let $\mathcal{X} = \{0, 1\}^T$, and let P_{par} be the probability mass function on $\mathcal{X}$ under which the first $T - 1$ bits are independent fair coins and the last bit is their parity. Let Q_{coin} be the probability mass function on $\mathcal{X}$ under which all T bits are independent fair coins. For an integer $n \geq 1$, a probability mass function Q on $\mathcal{X}$, and a sample $x^{(1)}, \dots, x^{(n)} \in \mathcal{X}$, define its empirical log loss per token by

$$\hat{L}_n(Q) = -\frac{1}{nT} \sum_{i=1}^n \log Q(x^{(i)}).$$

(a) For the model class $\{P_{\text{par}}, Q_{\text{coin}}\}$ and any sample from P_{par} , derive

$$\hat{L}_n(P_{\text{par}}) = \frac{T-1}{T} \log 2, \quad \hat{L}_n(Q_{\text{coin}}) = \log 2, \quad \hat{L}_n(Q_{\text{coin}}) - \hat{L}_n(P_{\text{par}}) = \frac{\log 2}{T}.$$

For $x_{1:t} \in \{0, 1\}^t$, let $\text{par}(x_{1:t}) = (\sum_{s=1}^t x_s) \bmod 2$. Verify that for every $x_{<T} \in \{0, 1\}^{T-1}$,

$$P_{\text{par}}(\text{par}(x_{<T}) \mid x_{<T}) = 1, \quad Q_{\text{coin}}(\text{par}(x_{<T}) \mid x_{<T}) = \frac{1}{2}.$$

(b) Now use the class of all independent-bit laws $Q = \bigotimes_{t=1}^T \text{Bernoulli}(q_t)$ on $\mathcal{X}$, where $q_t \in [0, 1]$ for every $t \in [T]$. Derive the population sequence log loss

$$\mathbb{E}_{P_{\text{par}}}[-\log Q(X_{1:T})] = \sum_{t=1}^T \left[-\frac{1}{2} \log q_t - \frac{1}{2} \log(1 - q_t) \right]$$

and show that its unique minimizer is $q_t = 1/2$ for every t .

Solution

Write your solution here.

Points: (a): 6, (b): 8, (c): 6; total 20.

Exercise 3: Latent likelihood and the forward algorithm.

⦿ Derive the forward recursion and explain why it replaces enumeration of d^{T+1} hidden trajectories by $O(Td^2)$ computation.

Let d, N, T be positive integers. Let $\pi \in [0, 1]^d$ be a probability vector, and let $A \in [0, 1]^{d \times d}$ and $C \in [0, 1]^{N \times d}$ satisfy

$$\sum_{i=1}^d A_{ij} = 1, \quad \sum_{a=1}^N C_{aj} = 1 \quad \text{for every } j \in [d].$$

Let Q be the joint probability mass function of a hidden Markov model on

$$[d]^{T+1} \times [N]^T.$$

The initial-state pmf is π . Given previous state j , the next-state pmf is the column $A_{\cdot j}$. Given current state j , the emission pmf is the column $C_{\cdot j}$. These conditional pmfs do not depend on t . Fix $x_{1:T} \in [N]^T$. For $s_{0:T} \in [d]^{T+1}$, write $Q(s_{0:T}, x_{1:T})$ for the joint mass and $Q(x_{1:T})$ for the marginal mass of $x_{1:T}$.

(a) Derive

$$Q(s_{0:T}, x_{1:T}) = \pi_{s_0} \prod_{t=1}^T A_{s_t s_{t-1}} C_{x_t s_t}, \quad Q(x_{1:T}) = \sum_{s_{0:T} \in [d]^{T+1}} \pi_{s_0} \prod_{t=1}^T A_{s_t s_{t-1}} C_{x_t s_t}.$$

Verify that the sum has d^{T+1} terms.

(b) For $t \in [T]$, let Q_t be the joint marginal pmf on $[N]^t \times [d]$. Define $\Lambda_0 = \pi$ and $\Lambda_t \in \mathbb{R}^d$ by $\Lambda_t(j) = Q_t(x_{1:t}, j)$ for $j \in [d]$. Derive a recursion for Λ_t and show that, in vector form, it is

$$\Lambda_t = \text{diag}(C_{x_t, 1}, \dots, C_{x_t, d}) A \Lambda_{t-1}.$$

(c) Let $\mathbf{1} \in \mathbb{R}^d$ be the all-ones vector. Show that $Q(x_{1:T}) = \mathbf{1}^\top \Lambda_T$ and that the recursion evaluates it in $O(Td^2)$ arithmetic operations, rather than summing d^{T+1} terms.

Solution

Write your solution here.

Points: (a): 14, (b): 16; total 30.

Exercise 4: Permanent state and finite-window prediction.

⦿ Quantify the prediction loss from a finite context when information is stored in a permanent latent state.

Let $n \geq 1$ be an integer. On a common probability space, let U and $(W_t)_{t \geq 1}$ be real-valued random variables satisfying

$$U \sim \mathcal{N}(0, \tau^2), \quad Z_t = U + W_t, \quad W_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2),$$

where U is unobserved, is independent of the noise sequence, and $\tau, \sigma > 0$. The goal is to predict Z_{n+1} given $Z_{1:n} = z_{1:n}$, where $z_{1:n} \in \mathbb{R}^n$. For $k \geq 1$, define

$$v_k = (\tau^{-2} + k\sigma^{-2})^{-1}.$$

(a) Show that

$$\mathcal{L}(U \mid Z_{1:n} = z_{1:n}) = \mathcal{N}(m_n, v_n), \quad m_n = v_n \sigma^{-2} \sum_{i=1}^n z_i.$$

Starting from $a_0 = 0$, derive the running-sum recursion

$$a_k = a_{k-1} + z_k, \quad m_k = \frac{a_k}{\sigma^2/\tau^2 + k}.$$

(b) Let $1 \leq p \leq n$. Consider a full-history predictor $h_n(Z_{1:n})$ and a window predictor $h_p(Z_{n-p+1:n})$, where h_n and h_p are arbitrary prediction functions. Show that

$$\min_{h_n} \mathbb{E}[(Z_{n+1} - h_n(Z_{1:n}))^2] = \sigma^2 + v_n$$

and

$$\min_{h_p} \mathbb{E}[(Z_{n+1} - h_p(Z_{n-p+1:n}))^2] = \sigma^2 + v_p.$$

Show that the minima are attained by

$$\hat{Z}_{n+1}^{\text{full}} = \mathbb{E}[U \mid Z_{1:n}], \quad \hat{Z}_{n+1}^{\text{win}} = \mathbb{E}[U \mid Z_{n-p+1:n}].$$

Your result must apply to all predictors based on the indicated observations, including nonlinear predictors. For fixed p , show that the first minimum tends to σ^2 , while the second remains $\sigma^2 + v_p$, as $n \rightarrow \infty$.

(c) *Optional extension.* Let P_n and $Q_{p,n}$ be the following random probability laws on $\mathbb{R}$:

$$P_n = \mathcal{L}(Z_{n+1} \mid Z_{1:n}), \quad Q_{p,n} = \mathcal{L}(Z_{n+1} \mid Z_{n-p+1:n}).$$

Identify these two Gaussian predictive laws and show that

$$\mathbb{E}[D(P_n, Q_{p,n})] = \frac{1}{2} \log \frac{\sigma^2 + v_p}{\sigma^2 + v_n},$$

where the expectation is under the stated joint law of $Z_{1:n}$.

(d) *Optional extension.* Suppose now that τ^2 is unknown, let $K \geq 1$ be an integer, and suppose that we observe K independent trajectories

$$Z_{j,t} = U_j + W_{j,t}, \quad j \in [K], \quad t \in [n],$$

where the U_j are iid $\mathcal{N}(0, \tau^2)$, the $W_{j,t}$ are iid $\mathcal{N}(0, \sigma^2)$, and the two collections are independent. Let $\bar{Z}_{j,n} = n^{-1} \sum_{t=1}^n Z_{j,t}$. Show that

$$\hat{\tau}_{K,n}^2 = \frac{1}{K} \sum_{j=1}^K \bar{Z}_{j,n}^2 - \frac{\sigma^2}{n}$$

is unbiased for τ^2 and, for fixed n , satisfies $\hat{\tau}_{K,n}^2 \rightarrow \tau^2$ almost surely as $K \rightarrow \infty$. Then set $K = 1$ and show that $\hat{\tau}_{1,n}^2 \rightarrow U_1^2$ almost surely as $n \rightarrow \infty$. Explain why this limit does not consistently estimate τ^2 .

HINTS For (b), let X be any random element (e.g., a vector), and let Y, Z be real-valued random variables satisfying $\mathbb{E}[Y^2], \mathbb{E}[Z^2] < \infty$. If Z is a function of X , then

$$\mathbb{E}[(Y - \mathbb{E}[Y | X])Z] = 0.$$

For (c), recall or derive that, for $a, b \in \mathbb{R}$ and $A, B > 0$,

$$D(\mathcal{N}(a, A), \mathcal{N}(b, B)) = \frac{1}{2} \left[\log \frac{B}{A} + \frac{A + (a - b)^2}{B} - 1 \right].$$

The identity in the hint for part (b) may also be useful.

Solution

Write your solution here.

Points: (a): 10, (b): 10; total 20.

Exercise 5: Dependence created by overlapping windows.

🔗 Compute the beta-mixing coefficient induced by overlapping iid windows and distinguish token independence from window independence.

Let $\mathcal{Z}$ be a finite or countable alphabet, let $p : \mathcal{Z} \rightarrow [0, 1]$ satisfy $\sum_{z \in \mathcal{Z}} p(z) = 1$, and let $(Z_t)_{t \geq 1}$ be iid $\mathcal{Z}$ -valued random variables with law p . Let $L \geq 2$ be an integer, and define the $\mathcal{Z}^L$ -valued windows $W_t = (Z_t, \dots, Z_{t+L-1})$. For an integer $a \geq 1$, define the beta-mixing coefficient

$$\beta_W(a) = \sup_{t \geq 1} \|\mathcal{L}(W_{1:t}, W_{t+a:\infty}) - \mathcal{L}(W_{1:t}) \otimes \mathcal{L}(W_{t+a:\infty})\|_{\text{TV}},$$

- (a) Show that the window process is $(L - 1)$ -dependent: collections of windows separated by at least L starting positions are independent. Deduce that $\beta_W(a) = 0$ for $a \geq L$.
- (b) For $1 \leq a < L$, show that

$$\beta_W(a) = 1 - \left(\sum_{z \in \mathcal{Z}} p(z)^2 \right)^{L-a}.$$

Explain from this formula why the coefficient is close to one when the collision probability $\sum_{z \in \mathcal{Z}} p(z)^2$ is small or when the number $L - a$ of shared positions is large, even though the underlying token process is independent.

HINTS For (b), identify the underlying variables that occur in both window collections.

Solution

Write your solution here.